

hopfield-llm

Energy-based hallucination detection for LLMs using Modern Hopfield networks. Extracts MLP memory banks, computes Hopfield retrieval energy and entropy over hidden states during prefill and generation, and produces layer-wise divergence metrics and scalar hallucination scores.

Quick Start (Local)

```
micromamba create -n hopfield-llm python=3.12 -y && micromamba activate hopfield-llm
pip install -r requirements.txt && pip install -e .

# Run stages individually
hopfield-llm build-banks --model qwen25_3b --output banks.pt
hopfield-llm run-trajectory --model qwen25_3b --dataset truthfulqa --banks banks.pt --output traj/
hopfield-llm analyze --trajectories traj/ --output analysis.json
hopfield-llm visualize --analysis analysis.json --output plots/

# Or run everything from a config
hopfield-llm run-experiment --config configs/experiments/exp01_truthfulqa_baseline.yaml
```

Quick Start (HPC)

```
bash setup_hpc.sh
./jobs/submit_pipeline.sh # defaults
MODEL=qwen25_7b DATASET=nq NUM_SHARDS=16 \
./jobs/submit_pipeline.sh # larger run
```

Pipeline Stages

Stage	Command	GPU	Input	Output
1	build-banks	Yes	model alias	banks.pt (MLP W_down weights)
2	run-trajectory	Yes	banks + dataset	.npz + .json per sample
3	analyze	No	trajectory dir	analysis.json (divergences, scores)
4	visualize	No	analysis JSON	plots/ (PNG layer-metric plots)

Each stage runs independently. `run-experiment` orchestrates all 4 from a YAML config with full experiment tracking (config snapshot, git info, timing, GPU metadata).

Supported Models

Alias	Model	Params	VRAM (4-bit)	Local (RTX 2050)	HPC (A100)
qwen25_1_5b	Qwen2.5-1.5B-Instruct	1.5B	~1.2 GB	Yes	Yes
qwen25_3b	Qwen2.5-3B-Instruct	3.0B	~2.0 GB	Yes	Yes
llama32_3b	Llama-3.2-3B-Instruct	3.2B	~2.1 GB	Tight	Yes
phi3_mini	Phi-3-mini-4k-instruct	3.8B	~2.4 GB	Marginal	Yes
gemma2_2b	gemma-2-2b-it	2.5B	~1.8 GB	Yes	Yes
qwen25_7b	Qwen2.5-7B-Instruct	7.6B	~4.5 GB	No	Yes
llama31_8b	Llama-3.1-8B-Instruct	8.0B	~5.0 GB	No	Yes
gemma2_9b	gemma-2-9b-it	9.2B	~5.5 GB	No	Yes
mistral7b	Mistral-7B-Instruct-v0.3	7.2B	fp16 only	No	A100-20GB+

Supported Datasets

- **TruthfulQA** ([truthfulqa](#)) — 817 questions testing factual accuracy
- **TriviaQA** ([triviaqa](#)) — Trivia questions with multiple answer aliases
- **Natural Questions** ([nq](#)) — Google search questions with short answers

Configuration

Experiment parameters are defined in YAML configs ([configs/experiments/](#)):

```

experiment:
  name: exp01_truthfulqa_baseline
model:
  alias: qwen25_3b
  load_in_4bit: true
dataset:
  name: truthfulqa
  max_samples: null
  seed: 42
trajectory:
  bank: down          # gate | up | down | gate_plus_up | gate_up_concat
  beta: 15.0
  energy_mode: dot    # dot | cosine
analysis:
  score_metric: js     # js | kl_fwd | hellinger | delta_energy | ...
  score_aggregation: mean
output:
  base_dir: outputs

```

Environment configs ([configs/environments/](#)) auto-detect local vs HPC via [HOPFIELD_ENV](#) or [SLURM_JOB_ID](#).

Repository Layout

src/hopfield_llm/	Python package
cli/	CLI entry point (5 subcommands)
models/	HFLLM loader + architecture resolution + model
profiles	
hooks/	Forward-hook activation capture (prefill + generation)
memory/	Memory bank construction from MLP weights
queries/	Query extraction strategies (swappable)
metrics/	Energy computation, divergences, hallucination scoring
datasets/	BaseDataset + TruthfulQA, TriviaQA, NQ adapters
pipeline/	Stage functions, analysis, ExperimentConfig
storage/	Artifact save/load (torch + JSON)
utils/	Logging, environment config, experiment tracking
visualization/	Matplotlib plots from analysis JSON
configs/	
environments/	local.yaml (RTX 2050) + hpc.yaml (A100)
experiments/	Experiment YAML configs
jobs/	SLURM templates + submission script
scripts/	Local convenience scripts + result inspection
outputs/	Experiment outputs (tracked run directories,
gitignored)	
tests/	
unit/	Pure function tests
integration/	Pipeline integration tests
docs/	Architecture, setup, and usage documentation

Documentation

- [Architecture](#) — Module responsibilities, data flow, design decisions
- [Setup](#) — Installation for local and HPC environments
- [Usage](#) — CLI, Python API, adding datasets, query extractors